



INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

TD N° 4– Administration Systèmes

Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

Date: 29/03/2026

L'objectif de ce TD est de :

- analyser les ports ouverts
- configurer un firewall simple
- écrire des règles de filtrage réseau
- comprendre les chaînes de filtrage
- comparer différentes solutions de firewall Linux
- analyser une configuration SSH
- sécuriser les accès
- mettre en place l'authentification par clé
- limiter les connexions SSH
- appliquer les bonnes pratiques de sécurité

1. Analyse du serveur

```
bribri@bribri1: ~  
https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.  
1 additional security update can be applied with ESM Apps.  
Learn more about enabling ESM Apps service at https://ubuntu.com/esm  
  
bribri@bribri1:~$ ss -tuln  
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process  
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:*  
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:*  
udp UNCONN 0 0 192.168.1.27%enp0s3:68 0.0.0.0:*  
udp UNCONN 0 0 0.0.0.0:111 0.0.0.0:*  
udp UNCONN 0 0 :::111 :::*  
udp UNCONN 0 0 [fe80::a00:27ff:fe07:48de]%enp0s3:546 :::*  
tcp LISTEN 0 4096 0.0.0.0:111 0.0.0.0:*  
tcp LISTEN 0 511 0.0.0.0:80 0.0.0.0:*  
tcp LISTEN 0 100 0.0.0.0:25 0.0.0.0:*  
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:*  
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:*  
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:*  
tcp LISTEN 0 4096 :::111 :::*  
tcp LISTEN 0 511 :::80 :::*  
tcp LISTEN 0 100 :::25 :::*  
tcp LISTEN 0 4096 :::22 :::*  
bribri@bribri1:~$
```

```
bribri@bribri1: ~  
[2]+ Stopped systemctl list-units --type=service  
bribri@bribri1:~$ systemctl list-units --type=service  
UNIT LOAD ACTIVE SUB DESCRIPTION  
apache2.service loaded active running The Apache HTTP Server  
apparmor.service loaded active exited Load AppArmor profiles  
apport.service loaded active exited automatic crash report generation  
blk-availability.service loaded active exited Availability of block devices  
console-setup.service loaded active exited Set console font and keymap  
cron.service loaded active running Regular background program processing daemon  
dbus.service loaded active running D-Bus System Message Bus  
finalrd.service loaded active exited Create final runtime dir for shutdown pivot root  
getty@tty1.service loaded active running Getty on tty1  
keyboard-setup.service loaded active exited Set the console keyboard layout  
kmod-static-nodes.service loaded active exited Create List of Static Device Nodes  
lvm2-monitor.service loaded active exited Monitoring of LVM2 mirrors, snapshots etc. using dmeventd or progress polling  
ModemManager.service loaded active running Modem Manager  
multipathd.service loaded active running Device-Mapper Multipath Device Controller  
networkd-dispatcher.service loaded active running Dispatcher daemon for systemd-networkd  
plymouth-quit-wait.service loaded active exited Hold until boot process finishes up  
plymouth-quit.service loaded active exited Terminate Plymouth Boot Screen  
plymouth-read-write.service loaded active exited Tell Plymouth To Write Out Runtime Data  
plymouth-start.service loaded active exited Show Plymouth Boot Screen  
polkit.service loaded active running Authorization Manager  
postfix.service loaded active exited Postfix Mail Transport Agent  
postfix@-.service loaded active running Postfix Mail Transport Agent (instance -)  
rpcbind.service loaded active running RPC bind portmap service  
rsyslog.service loaded active running System Logging Service  
setvtrgb.service loaded active exited Set console scheme  
snapd.apparmor.service loaded active exited Load AppArmor profiles managed internally by snapd  
snapd.seeded.service loaded active exited Wait until snapd is fully seeded  
ssh.service loaded active running OpenBSD Secure Shell server  
sysstat.service loaded active exited Resets System Activity Logs  
systemd-binfmt.service loaded active exited Set Up Additional Binary Formats  
systemd-journal-flush.service loaded active exited Flush Journal to Persistent Storage  
systemd-journald.service loaded active running Journal Service  
systemd-logind.service loaded active running User Login Management
```

2. Firewall simple avec UFW

```

bribri@bribri1:~$ sudo apt install ufw -y
[sudo] password for bribri:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
bribri@bribri1:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
bribri@bribri1:~$ sudo ufw status
Status: active
bribri@bribri1:~$ sudo ufw status
Status: active
bribri@bribri1:~$

```

```

bribri@bribri1: ~
plymouth-read-write.service      loaded active exited Tell Plymouth To Write Out Runtime Data
plymouth-start.service          loaded active exited Show Plymouth Boot Screen
polkit.service                  loaded active running Authorization Manager
postfix.service                 loaded active exited Postfix Mail Transport Agent
postfix@-service                loaded active running Postfix Mail Transport Agent (instance -)
rpcbind.service                loaded active running RPC bind portmap service
rsyslog.service                 loaded active running System Logging Service
setvtrgb.service               loaded active exited Set console scheme
snapd.apparmor.service          loaded active exited Load AppArmor profiles managed internally by snapd
snapd.seeded.service            loaded active exited Wait until snapd is fully seeded
ssh.service                     loaded active running OpenBSD Secure Shell server
sysstat.service                 loaded active exited Resets System Activity Logs
systemd-binfmt.service          loaded active exited Set Up Additional Binary Formats

[4]+  Stopped                  systemctl list-units --type=service
bribri@bribri1:~$ sudo ufw status
Status: active
bribri@bribri1:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.
bribri@bribri1:~$ sudo ufw status
Status: active
bribri@bribri1:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
bribri@bribri1:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
bribri@bribri1:~$

```

Autoriser les services necessaires

```

bribri@bribri1:~$ sudo ufw allow 22
Rule added
Rule added (v6)
bribri@bribri1:~$ sudo ufw allow 80
Rule added
Rule added (v6)
bribri@bribri1:~$ sudo ufw allow 443
Rule added
Rule added (v6)
bribri@bribri1:~$

```

Activer le firewall

```

bribri@bribri1:~$ sudo ufw enable
[sudo] password for bribri:
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
bribri@bribri1:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
bribri@bribri1:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)

bribri@bribri1:~$

```

3. Configuration avec iptables

```

bribri@bribri1:~$ sudo iptables -L
Chain INPUT (policy DROP)
target prot opt source destination
ufw-before-logging-input all -- anywhere anywhere
ufw-before-input all -- anywhere anywhere
ufw-after-input all -- anywhere anywhere
ufw-after-logging-input all -- anywhere anywhere
ufw-reject-input all -- anywhere anywhere
ufw-track-input all -- anywhere anywhere

Chain FORWARD (policy DROP)
target prot opt source destination
ufw-before-logging-forward all -- anywhere anywhere
ufw-before-forward all -- anywhere anywhere
ufw-after-forward all -- anywhere anywhere
ufw-after-logging-forward all -- anywhere anywhere
ufw-reject-forward all -- anywhere anywhere
ufw-track-forward all -- anywhere anywhere

Chain OUTPUT (policy ACCEPT)
target prot opt source destination
ufw-before-logging-output all -- anywhere anywhere
ufw-before-output all -- anywhere anywhere
ufw-after-output all -- anywhere anywhere
ufw-after-logging-output all -- anywhere anywhere
ufw-reject-output all -- anywhere anywhere
ufw-track-output all -- anywhere anywhere

Chain ufw-after-forward (1 references)

```

```
bribri@bribri1: ~  
Chain ufw-user-limit-accept (0 references)  
target      prot opt source      destination  
ACCEPT      all  --  anywhere    anywhere  
  
Chain ufw-user-logging-forward (0 references)  
target      prot opt source      destination  
  
Chain ufw-user-logging-input (0 references)  
target      prot opt source      destination  
  
Chain ufw-user-logging-output (0 references)  
target      prot opt source      destination  
  
Chain ufw-user-output (1 references)  
target      prot opt source      destination  
bribri@bribri1:~$ sudo iptables -P INPUT DROP  
bribri@bribri1:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACC  
iptables v1.8.10 (nf_tables): Chain 'ACC' does not exist  
Try 'iptables -h' or 'iptables --help' for more information.  
bribri@bribri1:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACC  
iptables v1.8.10 (nf_tables): Chain 'ACC' does not exist  
Try 'iptables -h' or 'iptables --help' for more information.  
bribri@bribri1:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT  
bribri@bribri1:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT  
bribri@bribri1:~$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT  
bribri@bribri1:~$ sudo iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT  
bribri@bribri1:~$
```

4. Firewall moderne avec nftables

```
bribri@bribri1:~$ sudo apt install nftables -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
nftables is already the newest version (1.0.9-1ubuntu0.1).  
nftables set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 17 not upgraded.  
bribri@bribri1:~$ sudo systemctl enable nftables  
Created symlink /etc/systemd/system/sysinit.target.wants/nftables.service → /usr/lib/systemd/system/nftables.service.  
bribri@bribri1:~$ sudo nano /etc/nftables.conf
```

5. Test de securite

```
bribri@bribri1:~$ nmap localhost  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-30 00:03 UTC  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0015s latency).  
Other addresses for localhost (not scanned): ::1  
Not shown: 996 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
25/tcp    open  smtp  
80/tcp    open  http  
111/tcp   open  rpcbind  
  
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds  
bribri@bribri1:~$
```

6-Analyse de la configuration actuelle

```

bribri@bribri1:~$ sudo cat /etc/ssh/sshd_config

# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key

```

7-Desactiver la connexion root

```

GNU nano 7.2 /etc/ssh/sshd_config

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
PermitRootLogin no
PasswordAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

|# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts

```

```

bribri@bribri1:~$ sudo systemctl restart ssh
bribri@bribri1:~$ |

```

8-Authentification par clé SSH

```

bribri@bribri1:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/bribri/.ssh/id_ed25519): ssh-copy-id bribri@192.168.2.26
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in ssh-copy-id bribri@192.168.2.26
Your public key has been saved in ssh-copy-id bribri@192.168.2.26.pub
The key fingerprint is:
SHA256:wjuw60mQYetngwrVdu90xCLVc0Cc0IMMghTb0LxQ3ro bribri@bribri1
The key's randomart image is:
+--[ED25519 256]--+
| .+o. .o. .+=o. |
| +.= . o +.oo |
| . 0 o . o |
| o 0 . o . |
| B + + S + |
| o = + + o |
| . E * o o . |
| .. + + + . |
| .+ . |
+-----[SHA256]-----+

```

```

bribri@bribri1:~$ ssh bribri@192.168.2.26
The authenticity of host '192.168.2.26 (192.168.2.26)' can't be established.
ED25519 key fingerprint is SHA256:8T5QCZWX/L5QF0rsZ9dK91M/2qx143LFrSgh2LQX3Yw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.2.26' (ED25519) to the list of known hosts.
bribri@192.168.2.26's password:
Permission denied, please try again.
bribri@192.168.2.26's password:
Permission denied, please try again.
bribri@192.168.2.26's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Mar 30 12:15:40 AM UTC 2026

System load:            0.03
Usage of /:              12.3% of 24.44GB
Memory usage:           7%
Swap usage:             0%
Processes:              128
Users logged in:        1
IPv4 address for enp0s3: 192.168.2.26
IPv6 address for enp0s3: fd5b:16fa:e40f:30:a00:27ff:fe07:48de
IPv6 address for enp0s3: 2803:9810:a0e1:9530:a00:27ff:fe07:48de

```

9-Desactiver l'authentification par mot de passe

```
GNU nano 7.2 /etc/ssh/sshd_config *
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
PermitRootLogin no
PasswordAuthentication no

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
|

bribri@bribri1:~$ sudo systemctl restart ssh
bribri@bribri1:~$ |
```

10-Changer le port SSH

```
GNU nano 7.2 /etc/ssh/sshd_config *
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

# When systemd socket activation is used (the default), the socket
# configuration must be re-generated after changing Port, AddressFamily, or
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none
|
```

```
bribri@bribri1:~$ sudo nano /etc/ssh/sshd_config
bribri@bribri1:~$ sudo systemctl restart ssh
bribri@bribri1:~$ ssh -p 2222 bribri@192.168.2.26
ssh: connect to host 192.168.2.26 port 2222: Connection refused
bribri@bribri1:~$ |
```

11-Protection contre brute force

```
bribri@bribri1:~$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-pyasyncore python3-pyinotify whois
Suggested packages:
  monit sqlite3 python-pyinotify-doc
The following NEW packages will be installed:
  fail2ban python3-pyasyncore python3-pyinotify whois
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 496 kB of archives.
After this operation, 2,572 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyasyncore all 1.0.2-2 [10.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 fail2ban all 1.0.2-3ubuntu0.1 [409 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-pyinotify all 0.9.6-2ubuntu1 [25.0 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 whois amd64 5.5.22 [51.7 kB]
Fetched 496 kB in 8s (59.9 kB/s)
Selecting previously unselected package python3-pyasyncore.
(Reading database ... 94089 files and directories currently installed.)
Preparing to unpack .../python3-pyasyncore_1.0.2-2_all.deb ...
Unpacking python3-pyasyncore (1.0.2-2) ...
Selecting previously unselected package fail2ban.
Preparing to unpack .../fail2ban_1.0.2-3ubuntu0.1_all.deb ...
Unpacking fail2ban (1.0.2-3ubuntu0.1) ...
Selecting previously unselected package python3-pyinotify.
Preparing to unpack .../python3-pyinotify_0.9.6-2ubuntu1_all.deb ...
Unpacking python3-pyinotify (0.9.6-2ubuntu1) ...
Selecting previously unselected package whois.
```

```
bribri@bribri1:~$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-30 00:21:47 UTC; 28s ago
     Docs: man:fail2ban(1)
    Main PID: 8149 (fail2ban-server)
      Tasks: 5 (limit: 4601)
    Memory: 26.6M (peak: 26.9M)
       CPU: 902ms
    CGroup: /system.slice/fail2ban.service
            └─8149 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Mar 30 00:21:47 bribri1 systemd[1]: Started fail2ban.service - Fail2Ban Service.
Mar 30 00:21:47 bribri1 fail2ban-server[8149]: 2026-03-30 00:21:47,276 fail2ban.configreader [8149]: WARNING 'allowip'
Mar 30 00:21:48 bribri1 fail2ban-server[8149]: Server ready
lines 1-14/14 (FND)
```

```
bribri@bribri1:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:  sshd
bribri@bribri1:~$ |
```

12-Test de sécurité

```

bribri@bribri1:~$ ss -tuln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0            127.0.0.54:53           0.0.0.0:*
udp        UNCONN     0           0            127.0.0.53%lo:53       0.0.0.0:*
udp        UNCONN     0           0          192.168.2.26%enp0s3:68  0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:111           0.0.0.0:*
udp        UNCONN     0           0            [::]:111              [::]:*
udp        UNCONN     0           0      [fe80::a00:27ff:fe07:48de]%enp0s3:546 [::]:*
tcp        LISTEN     0          4096            127.0.0.54:53           0.0.0.0:*
tcp        LISTEN     0          4096          127.0.0.53%lo:53       0.0.0.0:*
tcp        LISTEN     0          4096            0.0.0.0:111           0.0.0.0:*
tcp        LISTEN     0           100            0.0.0.0:25            0.0.0.0:*
tcp        LISTEN     0          4096            0.0.0.0:22            0.0.0.0:*
tcp        LISTEN     0          4096            [::]:111              [::]:*
tcp        LISTEN     0           511              *:80                  *:80
tcp        LISTEN     0           100            [::]:25              [::]:*
tcp        LISTEN     0          4096            [::]:22              [::]:*

bribri@bribri1:~$ nmap 192.168.2.26
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-30 00:25 UTC
Nmap scan report for bribri1 (192.168.2.26)
Host is up (0.00017s latency).
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
111/tcp   open  rpcbind

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
bribri@bribri1:~$

```

Commandes

```

bribri@bribri1:~$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
443 ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)

```

```

bribri@bribri1:~$ sudo iptables -L
Chain INPUT (policy DROP)
target prot opt source destination
ufw-before-logging-input all -- anywhere anywhere
ufw-before-input all -- anywhere anywhere
ufw-after-input all -- anywhere anywhere
ufw-after-logging-input all -- anywhere anywhere
ufw-reject-input all -- anywhere anywhere
ufw-track-input all -- anywhere anywhere

Chain FORWARD (policy DROP)
target prot opt source destination
ufw-before-logging-forward all -- anywhere anywhere
ufw-before-forward all -- anywhere anywhere
ufw-after-forward all -- anywhere anywhere
ufw-after-logging-forward all -- anywhere anywhere
ufw-reject-forward all -- anywhere anywhere
ufw-track-forward all -- anywhere anywhere

Chain OUTPUT (policy ACCEPT)
target prot opt source destination
ufw-before-logging-output all -- anywhere anywhere
ufw-before-output all -- anywhere anywhere
ufw-after-output all -- anywhere anywhere
ufw-after-logging-output all -- anywhere anywhere
ufw-reject-output all -- anywhere anywhere
ufw-track-output all -- anywhere anywhere

```

```
bribri@bribri1:~$ sudo nft list ruleset
# Warning: table ip filter is managed by iptables-nft, do not touch!
table ip filter {
    chain ufw-before-logging-input {
    }

    chain ufw-before-logging-output {
    }

    chain ufw-before-logging-forward {
    }

    chain ufw-before-input {
        iifname "lo" counter packets 24413 bytes 2049199 accept
        ct state related,established counter packets 17402 bytes 1350706 accept
        ct state invalid counter packets 0 bytes 0 jump ufw-logging-deny
        ct state invalid counter packets 0 bytes 0 drop
        ip protocol icmp icmp type destination-unreachable counter packets 0 bytes 0 accept
        ip protocol icmp icmp type time-exceeded counter packets 0 bytes 0 accept
        ip protocol icmp icmp type parameter-problem counter packets 0 bytes 0 accept
        ip protocol icmp icmp type echo-request counter packets 0 bytes 0 accept
        udp sport 67 udp dport 68 counter packets 0 bytes 0 accept
        counter packets 6 bytes 843 jump ufw-not-local
        ip daddr 224.0.0.251 udp dport 5353 counter packets 0 bytes 0 accept
        ip daddr 239.255.255.250 udp dport 1900 counter packets 0 bytes 0 accept
        counter packets 6 bytes 843 jump ufw-user-input
    }

    chain ufw-before-output {
```

```
bribri@bribri1:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-03-30 00:20:01 UTC; 7min ago
 TriggeredBy: ● ssh.socket
    Docs: man:sshd(8)
          man:sshd_config(5)
   Main PID: 7912 (sshd)
      Tasks: 1 (limit: 4601)
     Memory: 1.2M (peak: 2.5M)
        CPU: 87ms
    CGroup: /system.slice/ssh.service
            └─7912 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 30 00:20:01 bribri1 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 30 00:20:01 bribri1 sshd[7912]: Server listening on 0.0.0.0 port 22.
Mar 30 00:20:01 bribri1 sshd[7912]: Server listening on :: port 22.
Mar 30 00:20:01 bribri1 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

```
bribri@bribri1:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
`- Jail list:          sshd
bribri@bribri1:~$
```

Questions

1. Différence entre firewall stateful et stateless

R : Un pare-feu stateless traite chaque paquet indépendamment, sans tenir compte des connexions précédentes. Il applique des règles fixes basées sur l'adresse IP, le port ou le protocole, ce qui le rend simple et rapide mais limité en termes de sécurité. À l'inverse, un pare-feu stateful conserve une table d'états des connexions actives et peut déterminer si un paquet appartient à une session légitime déjà établie. Cette capacité d'analyse contextuelle permet de bloquer les paquets suspects

et de se protéger contre les attaques par usurpation de session. C'est pourquoi les firewalls stateful sont aujourd'hui la norme dans les environnements sécurisés .

2. Avantages de nftables par rapport à iptables

R : nftables est conçu pour remplacer iptables et apporte une architecture plus moderne et performante. Il unifie la gestion des protocoles (IPv4, Ipv6, ARP, bridge) dans une syntaxe unique, ce qui réduit la duplication des règles et simplifie l'administration. De plus, nftables améliore la vitesse d'évaluation des règles et facilite la gestion des configurations complexes. Sa syntaxe est plus lisible et flexible, ce qui en fait le choix recommandé sur les distributions Linux récentes. En somme, nftables est plus adapté aux environnements modernes où la scalabilité et la performance sont essentielles .

3. Pourquoi autoriser SSH avant d'activer un firewall

R : Avant d'activer un firewall, il est impératif d'autoriser le service SSH. En effet, la politique par défaut d'un firewall actif est généralement de bloquer tout trafic entrant. Si SSH n'est pas autorisé, l'administrateur perd immédiatement l'accès distant au serveur et doit intervenir physiquement pour le rétablir. Cela constitue une erreur critique d'administration et une violation du principe de continuité de service. Autoriser SSH avant l'activation garantit que l'administration distante reste possible et évite de « s'enfermer » hors du système .

4. Pourquoi désactiver la connexion root

R : Le compte root dispose de privilèges illimités, ce qui en fait une cible privilégiée des attaques par force brute. Sa désactivation en SSH repose sur plusieurs principes de sécurité : le principe du moindre privilège, qui limite les droits aux stricts besoins ; la réduction de la surface d'attaque, puisque root est universellement connu ; la traçabilité, car l'utilisation de sudo permet d'identifier quel utilisateur a exécuté une commande ; et enfin la limitation des dégâts, car la compromission d'un compte utilisateur standard n'entraîne pas automatiquement la compromission totale du système. Cette mesure est donc essentielle pour renforcer la sécurité et améliorer l'auditabilité .

5. Avantages de changer le port SSH

- **Réduction des attaques automatisées** : la majorité des bots ciblent le port 22 par défaut.
- **Moins de bruit dans les logs** : facilite la détection des activités suspectes.
- **Contournement des restrictions réseau** : utile si le port 22 est filtré.
- **Limite** : ce n'est pas une protection forte, mais une mesure complémentaire dans une stratégie de défense en profondeur.

6. Rôle de Fail2Ban

Fail2Ban est un système de prévention des intrusions (IPS) qui agit au niveau applicatif. Son fonctionnement repose sur trois étapes : l'analyse des journaux des services exposés (SSH,

Apache, etc.), la détection d'anomalies lorsqu'un seuil de tentatives échouées est atteint, et enfin la mise en place d'une contre-mesure automatique en bannissant l'adresse IP fautive via le firewall. Fail2Ban constitue ainsi une réponse active et automatisée aux attaques par force brute, réduisant significativement l'exposition des services aux tentatives répétées d'intrusion .

Conclusion

Ce TD avait pour objectif de mettre en pratique les bases de la sécurisation d'un serveur Linux : analyser les ports ouverts, configurer un firewall, durcir l'accès SSH et mettre en place des protections contre les attaques. Les réponses montrent que chaque mesure du choix d'un firewall moderne à la désactivation du compte root, en passant par l'autorisation préalable de SSH et l'usage de Fail2Ban participe à une stratégie de défense en profondeur.